CS 4644/7643: Deep Learning
Project Milestone Report & Rubric

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Discussions: `https://piazza.com/class/ms43b2tjxa63lq#`

Due: 4 November 2026, 11:59 pm ET

Logistics

1. **One PDF per group for all future submissions.**
2. **Where/format.** Submit a PDF on Gradescope (*Group Submission*; add all team members). Page limit: no more than **5 pages** for main text (references do not count). Appendix allowed only for figures; no additional text.
3. **Templates.** This year we are using the ICLR conference template. You can download it from Canvas. You are welcome to reuse content from your proposal if appropriate. While creating the report, add all team members as authors.

Purpose of the Milestone

The project milestone report serves the following purposes:

1. **Progress Update:** document the work you have completed since your initial project proposal.
2. **Problem Refinement:** clarify or refine your problem statement based on your research and initial experiments.
3. **Methodology Review:** describe the technical approach in detail, including any adjustments made since your proposal.
4. **Preliminary Results/Trials:** present some trials on baselines or your methods; if no results can be shown, show your current effort to implement baselines/your methods.
5. **Make a Plan:** create a concrete plan towards finishing your project.

Your Milestone Report Should Include the Following Sections

1. **Introduction**
   In this section, you need to describe the problem you want to solve. You may also include the following (you do not have to cover all of them):

   - the motivation for solving this problem
   - some key parts of the problem:
     Input/Output: The proposed method takes as input RGB images of dimension $640 \times 480 \times 3$. The output is a sequence of predicted waypoints representing the robot's future trajectory in SE(2), parameterised as (x, y, θ). **Training Conditions:** This includes the supervision strategy (i.e., the labels used and how they are obtained), the choice of network architecture (e.g., CNN, Transformer, ResNet), and the optimisation objective (loss functions and regularisation schemes)
   - your understanding of the problem

2. **Related Work**
   Summarise relevant literature and explain how it informs your approach (aim for at least 3 citations), and how your direction differs from or improves on previous works in resolving the remaining key challenges. Briefly describe the datasets you will use, if any, or data-collection methods if you are collecting your own data.

3. **Problem Decomposition & Detailed Technical Approach**

   - the precise formulation of your project as a deep learning problem (what are you training on? what are you trying to predict?)
   - the detailed structure of your tentative method (data processing, model/algorithm design, training details)
   - steps to evaluate the success of your chosen method (the evaluation procedure: splits/metrics)
   - **CS7643 Groups Only:** Three research hypotheses (H1-H3) to better understand your proposed method tentative method (see A for guidance). This exists to guide you towards designing insightful ablations/experiments and not just hyperparameter sweeps

4. **Experiments and Results**
   A baseline is a simple, well-justified method that fits your problem domain and sets a clear reference point. Choose 1–3 baselines (e.g., a classical heuristic, a lightweight supervised model, or a prior published approach). Explain how they work at a high-level and why it is appropriate for the problem. Evaluate them with the same data, metrics, and protocol you will use for your own method to ensure fair comparison; report numbers/visualizations.

   For the milestone, **baseline results are required** but results from your method are optional. In either case, document progress on your method(e.g., completed modules, pipeline status).

5. **Next Steps**
   Provide a concrete plan to finish: key milestones, rough timeline, potential challenges and mitigations, and how you will evaluate final success.

Grading Rubric for Groups *with* 7643 students (100 points)

Category & Criteria	Points
Formatting (8)	
Write 4-5 pages main-text (references excluded) and use the correct template; reasonable readability and structure.	8
Introduction (18)	
Clear statement of problem importance and goals	8
Scope/assumptions, inputs/outputs, training conditions	8
Include a teaser/draft figure	2
Related Work (12)	
A short survey of related works ($\geq$5 citations)	4
Demonstrates understanding and connection to your approach	8
Problem Decomposition & Detailed Technical Approach (32)	
Three hypotheses (H1–H3), specific and testable	8
Concrete technical approach for the stated problem	8
Sufficient technical detail to enable implementation	8
Evaluation procedure and concrete metrics	8
Experiments & Results (20)	
Established dataset or data-collection pipeline specified	4
Baseline evaluated with proposed metrics	8
Results for your method; or, in the absence of results, clear evidence of implementation effort (numbers/visuals preferred)	8
Next Steps (10)	
Plan to end-of-semester: milestones, rough timeline, potential challenges and mitigations, and success criteria	10
Total	**100**

Grading Rubric for Groups *without* 7643 Students (100 points)

Category & Criteria	Points
Formatting (8)	
Write 4-5 pages main-text (references excluded) and use the correct template; reasonable readability and structure.	8
Introduction (18)	
Clear statement of problem importance and goals	8
Scope/assumptions, inputs/outputs, training conditions	8
Include a teaser/draft figure	2
Related Work (16)	
A short survey of related works ($\geq$5 citations)	8
Demonstrates understanding and connection to your approach	8
Problem Decomposition & Detailed Technical Approach (24)	
Concrete technical approach for the stated problem	8
Sufficient technical detail to enable implementation	8
Evaluation procedure and concrete metrics	8
Experiments & Results (24)	
Established dataset or data-collection pipeline specified	8
Baseline evaluated with proposed metrics	8
Results for your method; or, in the absence of results, clear evidence of implementation effort (numbers/visuals preferred)	8
Next Steps (10)	
Plan to end-of-semester: milestones, rough timeline, potential challenges and mitigations, and success criteria	10
Total	**100**

A Examples of Good and Bad Hypotheses

A good research hypothesis should include both independent and dependent variables while easy to be tested via experiments. The hypothesis should explain what you expect to happen during experiments.

A.1 Good Hypotheses

- Applying domain randomization (textures, lighting, distractors) during training will *improve sim-to-real grasp success* compared with training without randomization, under identical data quantity and optimizer settings.
- For the same encoder and dataset, a cosine learning-rate schedule will *reach peak validation accuracy earlier* than a fixed learning rate, aggregated across multiple random seeds and identical batch sizes.
- Adding depth (RGBD) to RGB inputs will *increase collision-free navigation success* on unseen layouts relative to RGB-only, with architecture, compute, and training steps held constant.

A.2 Bad Hypotheses

- More data makes the robot more robust.
- With the same model, a cosine learning-rate schedule *converges faster* than a fixed learning rate.
- Adding depth will help navigation.